



उत्पाद मैनुअल

भा. मा. 17790: 2022 के अनुसार
घरेलु उपयोग के लिए उष्मारोधी फ्लास्क के लिए
दस्तावेज संख्या-पी एम/आई एस 17790/2/ फरवरी 2024

भारतीय मानक ब्यूरो की स्कीम-I (अनुरूपता मूल्यांकन) विनियम, 2018 के तहत यह उत्पाद मैनुअल प्रमाणीकरण के प्रचालन में रीति और पारिश्रिता की सुसंगतता सुनिश्चित करने के लिए सभी क्षेत्रीय/शाखा कार्यालयों और लाइसेंसी द्वारा संदर्भ सामग्री के रूप में उपयोग किया जाएगा। बीआईएस प्रमाणीकरण लाइसेंस/प्रमाणपत्र प्राप्त करने के इच्छुक भावी आवेदकों द्वारा भी इस दस्तावेज का उपयोग किया जा सकता है।

PRODUCT MANUAL FOR INSULATED FLASK FOR DOMESTIC USE ACCORDING to IS 17790:2022

Document No.-PM/IS 17790/2/ February 2024

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure coherence of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification licence/certificate.

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1.	Product	:	IS 17790: 2022
	Title	:	Insulated Flask For Domestic Use
	No. of Amendments	:	Nil
2.	Sampling Guidelines:		
a)	Raw material	:	• Welding electrode- Grade 308 or higher , of IS 5856 • Inner container- Grade 304 or higher of IS 5522 • Outer Container, if of steel, - N2 of IS 15997 or Grade 304 or higher of IS 5522 • Other components as per Cl. 7.1 of IS 17990: 2022 <i>Material of plastic shall be declared and conformity to declaration shall be ensured through test certificate of supplier</i> <i>Material of insulation shall be declared and conformity to declaration shall be ensured through test certificate of supplier</i>
b)	Grouping guidelines	:	Please refer ANNEX –A
c)	Sample Size	:	8 Nos.
d)	Sampling Guidelines during surveillance	:	In addition to the sample of product drawn during factory and market surveillance, sample shall also be drawn for independent testing of chemical composition of raw material of inner container. The same shall also be specifically mentioned in the test request.
3.	List of Test Equipment	:	Please refer ANNEX –B
4.	Scheme of Inspection and Testing	:	Please refer ANNEX –C
5.	Possible tests in a day	:	Capacity (Cl. 7.2), Fixing strength of Handle (Cl. 7.6), Strength of Shoulder strap (Cl. 7.7), Strength of Cord (Cl. 7.8), Leakage test (Cl. 7.9), Joint leakage test (Cl. 7.10), Checking for PUF (Cl. 7.12), Seepage (Cl. 7.13), Lever, operating device for operating or regulating flow test (Cl. 7.14), Length of Shoulder Strap (Cl. 7.15), Stability of flask (Cl. 7.16), Pout test (Cl. 7.17)

6.	Scope of the Licence:	:	Licence is granted to use Standard Mark as per IS 17790:2022 with the following scope:
	Name of product	:	Insulated Flask For Domestic Use
	Material of container	:	• Both inner and outer container is of stainless steel • Inner container is of stainless steel and outer container is of plastic
	Type of flask	:	Narrow mouth/Wide mouth
	Type of usage	:	Table top/ Use for transportation
	Nominal Capacity	:	From.....ml to Up to and including....ml

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ANNEX- A**Grouping Guidelines**

1. For the purpose of certification, Insulated flask for domestic use are classified as under:

Classification I: Type of Flask

- Wide Mouth above 45 mm
- Narrow Mouth 45 mm and below

Classification II: Nominal Capacities

Group A	Up to and including 1000 ml
Group B	Above 1000 ml

Classification III: Based on material

- Both inner and outer container is of stainless steel
- Inner container is of stainless steel and outer container is of plastic

Classification IV: Based on type of use

- Table top
 - Use for transportation
2. For covering all nominal capacities from a group in the scope of licence, the sample of the highest and lowest nominal capacities from each group (group A/B- capacity wise, as above) for each type of mouth portion and each types of usage, shall be tested separately for all requirements. Further, separate sample of each material type shall be tested separately to cover both types of materials as above.
 3. The Scope of Licence shall be restricted based on the Manufacturing and testing capabilities of the Manufacturer.
 4. During the operation of the licence, samples of all varieties covered in the scope of licence shall be drawn by rotation.

ANNEX B**List of Test Equipment**

Indicative list of major test equipment required to test as per the Indian Standard (For guidance only)

S.No.	Tests used in with Clause Reference	Test Equipment
1.	Minimum Thickness of Outer Container as per Cl 7.1.2.1, and of plastic casing as per Cl. 7.1.2.2	• Vernier Calipers or Micrometer
2.	Coating Thickness as per Cl 7.1.3	• Coating Thickness Gauge
3.	Capacity as per Cl 7.2	• Measuring Jars (with adequate marking to measure the capacity of flasks to the maximum capacity)
4.	Heat Retention Capability as per Cl 7.3	• Hot Plate • Thermometer • Air Conditioner • Stopwatch • Torque Wrench
5.	Cold Retention Capability as per Cl 7.4	• Refrigerator • Thermometer • Air Conditioner • Stopwatch • Torque Wrench
6.	Drop Impact Test as per Cl 7.5.1	• Drop Impact Test Fixture
7.	Pendulum Impact Test as per Cl 7.5.2	• Pendulum Impact Test Fixture
8.	Fixing Strength of the Handle as per Cl 7.6	• Test Fixture as per Figure 3 • Weighing Balance • Weights for adequate loading i.e. more than 6 times of mass of filled Flask • Stopwatch
9.	Test for Strength of Shoulder Strap as per Cl 7.7	• Test Fixture as per Figure 4 • Weighing Balance • Weights for adequate loading i.e. more than 10 times of mass of filled Bottle/Flask • Stopwatch
10.	Test for strength of cord as per Cl. 7.8	• Test Fixture • Steel balls of appropriate weight • Scale or vernier caliper to measure elongation

11.	Leakage Test as per Cl 7.9	• Measuring Jars • Hot Plate • Thermometer • Surfactant • Torque Wrench • Stopwatch
12.	Joint Leak Test as per Cl 7.10	• Weighing Balance (Accuracy 10mg) • Water Bath/Tub
13.	Staining Test as per Cl 7.11	• Suitable Soapy Solution • Acetone/Methylated Spirit • Soft Cloth • Timer/Stopwatch • Thermostatically Controlled Tank -2 Nos. • Weighing Balance • Glacial Acetic Acid (99 percent) • Sodium Chloride • Distilled Water
14.	Seepage test as per Cl. 7.13	• Blue indelible ink • Scale
15.	Test for Lever, device for operating or regulating flow test as per Cl. 7.14	• Suitable fixture with counter
16.	Test for length of the shoulder cap as per Cl. 7.15	• Scale
17.	Stability of Flask as per Cl 7.16	• Plane Inclined at 10 degrees
18.	Pour Test as per Cl 7.17	• Measuring Jars • Cup with mouth opening 60-80 mm diameter • White Paper • Tea/Coffee • Steel Scale

The above list is indicative only and may not be treated as exhaustive

ANNEX C**Scheme of Inspection and Testing****1. QUALITY ASSURANCE PLAN**

1.1 It is expected that manufacturers (licensees/applicants) will implement a Quality Assurance Plan i.e. a plan of regular testing and in-process controls, designed to ensure that the product bearing the Standard Mark conforms to all requirements of the Indian Standard.

1.2 The manufacturers shall define a Quality Assurance Plan defining the control unit i.e. Batch and the levels of control (i.e. the frequency and number of samples for conducting the different tests as per the Indian Standard) and submit the same to BIS Branch Office for information. The manufacturer shall comply with the same and maintain test records in accordance with para 2.4 below.

1.3 RECOMMENDED LEVELS OF CONTROL/CONTROL UNIT:

1.3.1 **Control Unit:** For the guidance of manufacturers, the recommended definition of Control Unit is quantity of all Flasks of same Type, Nominal Capacity and Material, manufactured by same method under similar conditions in a day.

1.3.2 For the guidance of manufacturers in preparing the Quality Assurance Plan, recommended levels of control are given in **Table 1**.

1.3.3 The manufacturer shall ensure inspection and testing as per the Quality Assurance Plan submitted by them on the whole production of the factory which is covered by this plan. Alternatively, the manufacturer has the option of adherence to the quality plan as per levels of control recommended in column 3 of Table 1.

1.4 However, all manufacturers shall ensure compliance of their products to all the requirements of the Indian Standard.

2. ENSURING COMPLIANCE THROUGH TESTING- It is expected that manufacturers (licensees/applicants) will establish a suitably equipped and staffed in-house laboratory (In house testing facility) for testing at least those parameters of the Indian Standard which require routine testing for ensuring quality of the product. This includes in-process controls as may be defined and put in place by the manufacturer, and testing parameters/requirements which can only be performed in the factory.

2.1 For the guidance of manufacturers, Table 1 giving the recommended levels of control is given below. Column 2 of Table 1 indicates routine tests where test equipment is required in-house as “R” or other tests which can be sub-contracted as “S”. Subcontracting is permitted to BIS recognized/empanelled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.2 For MSME manufacturers, the requirement of maintaining a laboratory/in-house testing facility for routine tests (indicated as “R” in Column 2 of Table 1) is also optional.

2.2.1 MSME manufacturers may utilize common cluster based facilities as per guidelines for the utilization of cluster based test facilities by MSMEs or the provisions of Sharing of testing facilities or get testing done from BIS recognized/empaneled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.3 Large Scale manufacturers shall maintain an in-house laboratory equipped at least with test facilities for routine tests (indicated as “R” in Column 2 of Table 1), where different tests given in the specification shall be carried out in accordance with the method given in the specification. They shall also implement a calibration plan for the in-house test equipment.

2.3.1 Alternatively, in lieu of an in-house laboratory, large scale manufacturers can also utilize the provisions of Sharing of testing facilities as per the Guidelines for Grant of Licence available on BIS website www.bis.gov.in. (Under Conformity Assessment>Product Certification Process). Even for subcontracted tests, provisions for sharing of testing facilities can be utilized.

2.4 TEST RECORDS- The manufacturers maintaining an in-house laboratory or utilizing common cluster based facilities or shared test facilities shall maintain test records for the tests carried out to establish conformity. For the tests being subcontracted to BIS recognized/empanelled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025, test reports issued by the laboratories shall be available for inspection by BIS.

3. LABELLING AND MARKING – As per the requirement of IS 17790:2022. The Standard Mark, as given in the Schedule of the licence, shall be marked on each Flask provided always that the product thus marked conforms to all the requirement of the specification. In addition, the licence no. CM/L-_____ and details of BIS website shall be marked at an appropriate place on product or package as follows: “For details of BIS certification please visit www.bis.gov.in”.

4. REJECTION - All the production which conforms to the Indian Standard and covered under scope of the licence shall be marked with Standard Mark. Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act,2016

TABLE 1

(1)				(2)	(3)		(4)
Test Details				Test equipment requirement R:required (or) S: Sub-contracting permitted	Levels of Control		Remarks
Clause	Requirements	Test Method			No. of Samples	Frequency	
		Clause	Reference				
5	Manufacture and Workmanship	5.1	IS 17790	R	Manufacturer shall exercise adequate levels of control to ensure that entire production should conform to the Standard. Appropriate records of testing indicating conformity shall be maintained		
5.3	Welding Defects	5.3	IS 17790	R	Each Flask		
5.3	Welding Electrode	5.3	IS 17790 IS 5856	S	One	Each Consignment	No further testing is required if material is ISI marked.
7.1	Material	7.1	IS 17790 IS 15997 IS 5522 IS 3565	S	One	Each Consignment	IS 15997 and IS 5522 are under Compulsory BIS Certification and it shall be ensured that only ISI Marked material are used and they are received along with Manufacturer’s test certificate. For other materials, No further testing is required in case received with test certificate or is ISI Marked. Material of plastic shall be declared and conformity to declaration shall be ensured through test certificate of supplier.

							Material of insulation shall be declared and conformity to declaration shall be ensured through test certificate of supplier.
7.1.2	Minimum Thickness of Outer Container	7.1.2	IS 17790	R	Three	Once in 3 months for every capacity	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.
7.1.3	Coating Thickness	7.1.3 7.2	IS 17790 IS 9730	R	Three	Once in 3 months	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.
7.1.3	Salt Water Corrosion Test	7.1.3 7.4	IS 17790 IS 9730	S	Three	Once in 3 months	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.
7.1.3	Adhesion Test	7.1.3 7.5	IS 17790 IS 9730	S	Three	Once in 3 months	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.

7.1.3	Lead Limit	7.1.3 Table 1	IS 17790 IS 9806	S	Three	Once in 6 months	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.
7.1.3	Cadmium Limit	7.1.3 Table 1	IS 17790 IS 9806	S	Three	Once in 6 months	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.
7.2	Capacity	7.2	IS 17790	R	Three	Each Control Unit	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.
7.3	Heat Retention Capability (1hour)	7.3 & Annex A	IS 17790	R	Three	Once in a month for each capacity and material	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.
7.3	Heat Retention Capability (6 hours & 12 hours)	7.3 & Annex A	IS 17790	R	Three	Once in a month for each capacity and material	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.

7.3	Heat Retention Capability (24 hours)	7.3 & Annex A	IS 17790	R	Three	Once in 3 months for each capacity and material	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass.
7.4	Cold Retention Capability	7.4 & Annex B	IS 17790	R	Three	Once in a month for each capacity and material	
7.5.1	Drop Impact	7.5.1	IS 17790	R	Three	Once in 3 months	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass
7.5.2	Pendulum Impact	7.5.2	IS 17790	R	Three	Once in 3 months	
7.6	Fixing Strength of Handle	7.6	IS 17790	R	Three	Once in 3 months	One sample of each capacity produced in 3 months shall be tested
7.7	Test for Strength of Shoulder Strap	7.7	IS 17790	R	Three	Once in 3 months	
7.8	Test for strength of cord	7.8	IS 17790	R	Three	Once in 3 months	
7.9	Leakage Test	7.9	IS 17790	R	Three	Each Control Unit	
7.10	Joint Leak Test	7.10	IS 17790	R	Three	Each Control Unit	
7.11	Staining Test	7.11	IS 17790	R	Three	Once in 3 months	

7.12	Checking of PUF insulation	7.12	IS 17790	R	One	Once in 3 months	--
7.13	Seepage test	7.13	IS 17790	R	Three	Once in 3 months	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass
7.14	Test for lever, operating device for operating or regulating flow test	7.14	IS 17790	R	Three	Once in 3 months	
7.15	Length of the Shoulder Strap	7.15	IS 17790	R	Three	Each Control Unit	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass
7.16	Stability of Flask	7.16	IS 17790	R	Three	Each Control Unit	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass
7.17	Pour Test	7.14	IS 17790	R	Three	Each Control Unit	In event of failure of even one of the 3 samples, additional 7 samples shall be tested and all of these shall pass